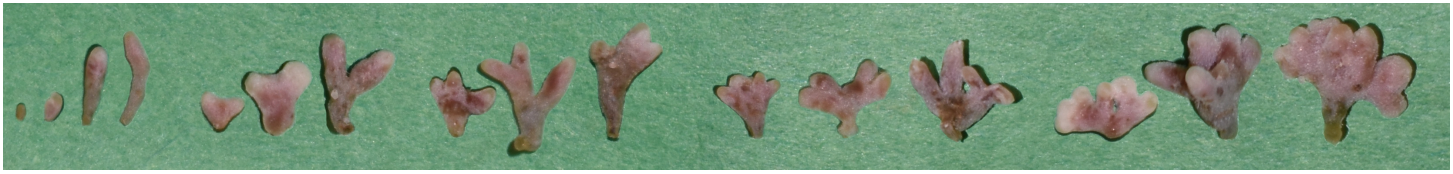
A) Branching variation in alfalfa



B) Experimental overview

1. Single-strain inoculation experiment

117 *S. meliloti* strains x 4 replicates

3 alfalfa varieties

Oneida (O) SW4107 (S) Vernal (V)

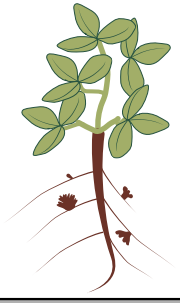
1500 plants grown for 9 months in a random block design

2. Rhizobia trait analysis

- 8 strain subset
- 6 nodules spanning size
- 4 replicates

C) Plant benefits

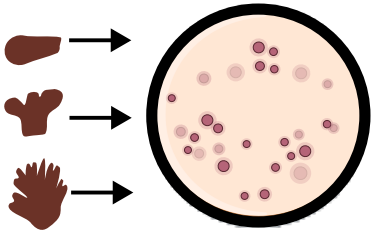
117 strain x 3 varieties - traits measured:

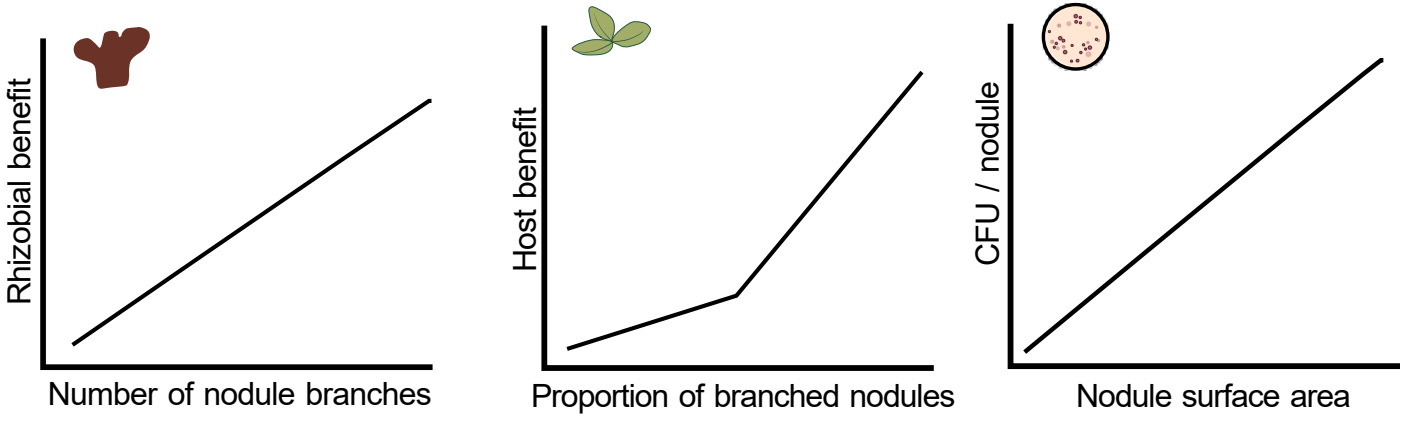


- Plant biomass
- Nodule number
- Number of branched nodules

D) Rhizobial benefits

8 strain subset - traits measued:



- # CFUs released per nodule
- Nodule Surface Area
- # of Nodule branches

E) Hypothesized relationships